SECTION - 13

JOINERY WORK

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SECTION - 13

JOINERY WORK

1.00 SCOPE

This specification covers the general requirements for wood work.

2.00 GENERAL REQUIREMENTS

2.01 The contractor shall furnish all materials, labour, operations, equipment, tools and plants, scaffolding and incidental necessary and required for the completion of all carpentry and joinery work as called for in the drawings and Bill of Quantities.

2.02 The carpentry and joinery work shall include the provision of fixing of fastening devices and hardware in accordance with the drawings and attached hardware schedule. Hardware and other materials will be supplied by the contractor after obtaining prior approval of the KEIPL/PM of samples for each item of hardware.

2.03 The Contractor shall strictly follow, at all stages of work, the stipulations contained in the Indian Standard Safety Code and the provisions of the Safety Rules as specified in the General Conditions of the Contract for ensuring safety of men and materials.

2.04 Any approval, instructions, permission, checking, review etc. whatsoever by the KEIPL/PM shall not relieve the Contractor of his responsibility and obligation regarding adequacy, correctness, completeness, safety, strength, workmanship etc.

3.00 CODES AND STANDARDS

3.01 All standards, specifications, acts, and codes of practice referred to herein shall be the latest editions including all applicable official amendments and revisions.

3.02 List of certain important Indian Standards, Acts and Codes applicable to this work is given below. However, the applicable standards and codes shall be as per but not limited to the list given below :

IS : 205	Non-ferrous metal butt hinges
IS : 281	Mild steel sliding door bolts for use with pad locks
IS : 287	Permissible moisture content for timber used for different purposes
IS : 303	Plywood for general purposes
IS : 848	Synthetic resin adhesives for plywood.
IS : 883	Code of practice for design of structural timber in building
IS : 1003	Timber panelled and glazed shutters
IS : 1141	Seasoning of timber - Code of practice

IS : 1328 Veneered decorative plywood.

IS : 1659 Block boards.

IS : 2202 Wooden flush door shutters

IS : 2835 Flat transparent sheet glass.

IS : 3087 Wooden particle boards (medium density) for general purposes

IS : 3097 Veneered particle boards.

IS : 3809 Fire resistant test for structures

IS : 4021 Timber door, window and ventilator frames

4.00 SAMPLES AND SHOP DRAWINGS

The contractor shall, before proceeding with the work, submit to the KEIPL/PMfor his approval complete samples of the various materials including hardware and fastening devices and shop drawings and large scale details covering all joinery works.

5.00 MATERIALS

5.01 **Timber :** Timber used shall be of type as stated in the Schedule and of best quality. All timber shall be of natural growth and uniform in texture and shall be well and properly seasoned. It shall be free from large, loose, dead, or cluster of knots, waves, injurious open shakes, borer holes, rot, decay, knots, discolouration, soft or spongy spots, hollow pockets with or centre heart and all other defects and blemishes. Timber shall conform to IS 883.

The seasoning of timber shall be as laid down in IS : 1141.

5.02 **Teak wood :** Teak should be from the forests of Dandeli and Mysore in Karnataka, Malabar in Kerala, Ballarshah in MP, Bulsar in Gujarat. It shall have uniform colour reasonably straight grains and shall be free from large loose dead knots cracks, shakes, warp, twists, bends bore holes, sap wood. For first class no individual hard and sound knot shall be more than 1 Sq.cm and the aggregate area of all knots shall not exceed 1/2% the area of piece and for second class 1.5 Sq.cm and 2% respectively. It shall be close grained and there shall not be less than 2 growth rings per cm width.

The wood shall be well seasoned. It shall be considered well seasoned if its moisture content does not exceed the limits as per IS 287. The moisture content shall be determined as per IS 287. Unless otherwise specified tolerance for door and window frames and shutters should not exceed provisions in IS 4021 & IS 1003 (part I & II).

a)	Frames	14%
b)	Planking, shutters etc.	12%

5.03 **Country wood & Malaysian Sal :** Country wood and Malaysian Sal are wood that are not classified as teak wood, honey wood, nandi and mathi wood. It shall have uniform colour, reasonably straight grains and shall be generally free from large loose dead knots, cracks, shakes, warp, twists, bends, bore holes, sap

wood. No individual hard and sound knot shall be more than 1.5 Sq.cm and the aggregate area of all knots shall not exceed 2% of the area of piece. It shall be close grained and there shall not be less than 2 rings per cm width.

The wood shall be well seasoned. It shall be considered well seasoned if its moisture content does not exceed the limits as per IS 287. The moisture content shall be determined as per IS 287. Unless otherwise specified tolerance for door and window frames and shutters should not exceed provisions in IS 4021 & 1003 - Part I & II.

5.04 **Laminated Particles Board:** Laminated particles board of approved shade bonded with phenol formaldehyde with all cut edges provided with teak lipping 12 mm thick with moisture proof coating with aluminium paint and chlorinated rubber paint along edges. The particle board shall conform to IS 3087.

5.05 **Veneered Particle Board :** Particle board bonded with Phenol Formaldehyde resin shall comply with the specification as per IS 3087 and shall have veneering on both sides with teak veneer faces. The exposed surface of veneered boards on both sides shall be polished fully.

5.06 **Glass:** All glass shall be of superior quality from approved manufacturer like Pilkington, Ashahi, St.Gobain, or approved equivalent, having uniform refractive index and free from flaws, speaks and bubbles. Sheet glass for glazing and framing purposes shall conform to IS : 2835 latest edition. The glass shall be brought to site in the original packing from the manufacturer and cut to size at site. The cut edges shall be straight and free from chips, spalls or any other damages. Wherever clear sheet glass is specified, it shall be flat drawn sheet glass and shall be atleast 4 mm thick or as specified. Where wire glass is mentioned, it shall be thick rolled glass with centrally embedded wire mesh or Georgian type. Where obscure glass is mentioned it shall have a cast surface on one side. The Contractor shall submit samples of glass for prior approval of the KEIPL / PM.

5.07 **Putty :** In general, the putty shall conform to IS : 420 latest edition. The putty shall be of best quality from approved manufacturer. It shall be brought to site in the manufacturer's original packing. Quick setting putty shall be used for windows and sashes except when glare reducing glass is used where it shall be of non-setting type.

6.00 DOORS, WINDOWS AND VENTILATORS

6.01 Doors, windows and ventilators etc. shall be in accordance with the drawing in every detail and all joiner's work shall be accurately set out, framed and finished in a proper workman like manner. Frames of doors, windows and ventilators, shutter styles and rails shall be of best solid wood of quality specified in the schedule of quantities. The scantlings shall be accurately planed smooth. Rebates, roundings and mouldings shall be made as shown on the drawings. Patching or plugging of any kind shall not be allowed.

Joints shall be simple, neat and strong. Framed joints shall be coated with suitable adhesive like glue or synthetic resin before the frames are put together. All mortice and tenon joints shall fit in fully and accurately without wedging or filling. The joints shall be pinned with hard wood or bamboo pins of 10 mm to 12 mm dia. or rust resisting star shaped metal pins of 8 mm diameter. All portions of timber abutting against or embedded in masonry or concrete shall be treated against termites by giving a coat of an approved wood preservative, for which no

extra cost will be paid. Putty shall not be used to cover any defects. Unless otherwise specified, all door frames shall have six holdfasts. Holdfasts shall be provided to the ventilators if directed. Size of holdfasts shall be 400 mm x 40 mm x 5 mm M.S. flat bent to shape with fish tail and shall be fixed to frame with sufficient number of screws as directed. When door/window frames are to be fixed to RCC column or RCC wall, holdfasts shall be substituted by suitable arrangements such as coach screws, rawl bolts/grip bolts etc. to secure frames to RCC column or RCC wall as directed by the KEIPL / PM. The frame shall be fixed only after getting the approval of the KEIPL / PM.

6.02 **Fire Resistence Door :** The fire resistance door one hour / two hours fire rated or as specified shall be provided. The fire rating to doors shall conform to PS 47648 and IS 3809 and as approved by the KEIPL / PM.

6.03 **Toilet Doors**

a) **External Doors :** Laminate shall be provided on inner face and decorative veneer or as per specification/ BOQ shall be provided on outer face of door shutters unless otherwise specified in the Schedule of probable itemsand all as per manufacturer specifications including necessary finishing and fittings etc.

b)**Internal doors :** Laminate shall be provided on inner face and on outer face of door shutters unless otherwise specified in the Schedule of probable itemsand all as per manufacturer specifications including necessary finishing and fittings etc.

6.04 **Frames :** These shall be cut and planed from first class timber of respective kind and of sizes as stated in the Schedule of Quantities or as detailed on the drawings, in the case of main frames as well as sub-frames.

6.05 **Architraves :** Architraves shall be cut and planed from best quality seasoned timber to sizes as shown in the drawings and fixed to the door or window frames with wood screws. Screw heads shall be sunk into frames. The side abutting the frame shall be finished with a coat of wood primer.

6.06 **Interior door shutters**

a) **Flush shutters :** Interior wood door shutters, unless otherwise noted or specified, shall be 38 / 32 mm thick phenol formal-dehyde synthetic resin thermo pressed flush shutters, teak veneered face on both faces as specified The shutters shall conform to IS:2202 (Part-I). Shutter make shall be approved by the KEIPL / PM.

The solid core shall be wood laminate prepared from battens of well seasoned and treated good quality wood having straight grains. The battens shall be of uniform size of about 2.5 cm width. These shall be properly glued and machine pressed together , with grains of each piece reversed from that of adjoining one. The Longitudinal joints of the battens shall be staggered and no piece shall be less than 50 cm. in length. Alternatively, the core shall be of solid teak particle board. Edges of the core shall be lipped internally with first class teak wood battens of 4 cm. minimum depth, glued and machine pressed along with the core.
The core surface shall then have two or three veneers firmly glued on each face. The first veneer (called cross band) shall be laid with its grains at right angles to those of the core and the second and the third

veneers with their grains parallel to those of the core. The under veneers shall be of good quality, durable and well seasoned wood. The face veneers shall be of minimum 1 mm thickness and of well matched and seasoned first class teak, laid along with grains of the core battens. The combined thickness of all the veneers on each face shall not be less than 4 mm. Thermosetting synthetic resin conforming to I.S. 303 or moisture-proof plywood grade M.P.F.I. shall be used in manufacture. All doors shall have external lipping alround 6 to 10 mm thick in addition to internal lipping.

Block boards and ply for various items of work called for shall be of approved make. Unless otherwise shown all block boards and ply shall be teak veneered faced on both faces.

Samples of flush doors, block boards, etc., shall be submitted to the KEIPL/PM for his approval and all shutters, etc., to be used in the work shall conform to the approved sample in all respects.

b) **Panelled Shutters :**

i) The thickness of the panels shall be 15 mm for 25 mm shutters and 20 mm for shutters of larger thickness. The edges of the panels shall be tapered symmetrically on all the sides for insertion into the styles and rails. The depth of rebate in frames for housing the shutters shall in all cases be 1.25 cm and the rebate in shutters, for closing, in double leaf shutter doors, and windows shall not be less than 2 cm. The edges of styles and rails all round the panels shall be moulded, rounded or chamfered on both faces of the shutter as directed by the KEIPL / PM.

ii) Factory made, kiln seasoned and chemically treated panelled shutters - The thickness and quality of wood of the shutters shall be as indicated in the item. Styles and rails of teak wood and panels made of 12 mm thick teak veneered particle board or similar material bonded with phenol formaldehyde synthetic resin adhesive conforming to IS : 3097 shall be used.

7.00 INSTALLATION

All timber for joinery work or wrought form work shall be of best Indian Teakwood unless otherwise specified and or specified quality, carefully selected free from sap and subject to inspection and approved before delivery. All teakwood to be used in this work shall be seasoned in an approved manner.

All workmanship pertaining to wood work shall be of best standard and all joinery work shall fit freely and without wedging or filling and of the kind directed by the KEIPL / PM. Wood work in frames shall be wrought and all frames joints shall be put together with white lead and pinned with hard wood pins. All exposed faces of joinery shall be thoroughly planed. Jointing shall be by means of mortice and tenon, dovetailed joints or as directed by the KEIPL / PM.

The joiner work shall be framed and put together immediately but not to be wedged up until required for fixing. The approval of the KEIPL/PMshall be obtained before any primer coat is applied.

The whole of joinery to be finished to the dimensions and sizes indicated in the drawings are subject to a tolerance to 1.50 mm of each planed face, but no allowance shall be given to flush doors, shutters, ply and other manufactured board etc.

All black boards etc. shall correspond to respective IS specifications. Flush doors shall correspond to relevant IS specifications. Samples of all black board, plywood, flush doors, etc. shall be got approved by the KEIPL/PMbefore placing order. All such items shall be of standard manufacture of reputed quality. Unless otherwise specified, all flush shutters shall be teak veneered on both sides. All flush shutters shall be of solid core and shall be provided with teakwood external lipping.

If after execution any shrinkage or bad workmanship is found, the Contractor shall forthwith replace or refix the same at his own cost, all as directed by the KEIPL / PM.
Rate quoted shall include for all framing, usual waste and cuttings and no extra rates or claims will be entertained on this account.

The rate for wood work shall include the cost of all sawing, planing, jointing, framing, labour and materials for raising and fixing and all workmanship and fixing and supplying of all strips, bolts, nails, trenails, spikes, keys, wedges, pins, screws etc. necessary for the framing as per specification and drawings. Edges of the beams, joists, posts, frames etc. shall be rounded, moulded or chamfered as directed without extra charges.

The Contractors shall be responsible to deliver all items at site of work. The Contractor will submit a programme of work in such a way that requirements for each floor commencing from ground floor is completed before the requirement of the next floor and arrange to have them fixed in position as the General Builders work progresses.

Timber in contact with masonry or concrete shall be treated with solignum paint or any approved anti-termite treatment before fixing.

The Contractor will be making necessary holes in concrete and masonry for fixing in position and grouting. The Contractor will be responsible for the proper fixing of partitions true to plumb and alignment until completion and grouting pockets with cement concrete and make good affected portions without claim to any extra.

The iron monger fittings shall be of heavy type cast brass oxidised and of approved manufacture. Samples of all iron monger items shall be got approved by the KEIPL/PM prior to procurement.

8.00 HARDWARE FITTINGS

All hardware fittings and fixtures shall be made with structural properties to sustain safety and withstand strains and stresses to which they are normally subjected to such as opening and closing, wind pressure etc. The fittings shall generally conform to relevant specifications.

They shall be made true, clean, straight, with sharply defined profiles and unless otherwise shown or specified, with true smooth surfaces and edges, free from defects, screw holes shall be counter sunk to suit the head of wood screws.

The metal shall be treated with finish as specified in the Bill of Quantities.

The approved hardware finish as specified in the Bill of Quantities.

9.00 **PRESERVATIVE TREATMENT**

All wood work in contact with masonry shall be painted with approved sphalt or bitumen paint before placing. Care shall be taken to keep exposed faces clear from tar, etc. Tarfelt shall be used to isolate wood from masonry wherever practically possible. All concealed wood members in ceiling, partitions, cabinet work, etc., shall be treated fully and liberally with solignum before placing in position.

10.00 **MODE OF MEASUREMENT**

Unless noted otherwise in the schedule of probable items, the method of measurement for various items shall be generally in accordance with IS 1200 subject to the following:

1) All work shall be measured net as fixed. No extra measurement will be given for shape, joints, splayed, meeting stiles of doors and windows and shall be measured in unit of square metre. Door frames shall be in cubic metre, if measured separately.

2) Area over one face inclusive of exposed frame thickness (excluding width of cover mould) shall be measured in case of doors, windows and ventilators when frames are included in the item. Portions embedded in masonry or flooring shall not be measured.

STEEL DOORS

1.00 **SCOPE**

1.01 This specification covers the design, supply of materials, fabrication and installation of factory made special type of approved make steel fire doors and general purpose doors with all accessories and including supply and installation of hardwares.

2.00 **GENERAL REQUIREMENTS**

2.01 The Contractor shall furnish all materials, labour, operations, equipment, tools & plant, scaffolding and incidentals necessary and required for the completion of all metal work in connection with steel doors, as called for in the drawings, specifications and Schedule of probable itemswhich cover the major requirements only. Anything called for in the tender documents shall be considered as applicable to the items of work concerned. The supply and installation of additional fastenings, accessory features and other items not specifically mentioned, but which are necessary to make a complete functioning installation shall form a part of this contract.

2.02 All metal work shall be free from defects, impairing strength, durability and appearance and shall be of the best quality for purposes specified made with structural properties to withstand safely strains, stresses to which they shall normally be subjected to.

2.03 All fittings shall be of high quality and as specified and as per approval of KEIPL/PM.

2.04 The Contractor shall strictly follow, at all stages of work, the stipulations contained in the Indian Standard Safety Code and the provisions of the safety code and the provision of the safety rules as specified in the General Conditions of the Contract for ensuring safety of men and materials.

2.05 Any approval, instructions, permission, checking, review, etc. whatsoever by the KEIPL/PM, shall not relieve the Contractor of his responsibility and obligation regarding adequacy, correctness, completeness, safety, strength, quality, workmanship, etc.

3.00 CODES AND STANDARDS

3.01 All standards, specifications, acts, and codes of practice referred to herein shall be the latest editions including all applicable official amendments and revisions.

3.02 List of certain important Indian Standards, Acts and Codes applicable to this work is given below. However, the applicable standards and codes shall be as per but not limited to the list given below :

IS : 277 Galvanised steel sheet (plain and corrugated)

IS : 3614 Metallic and non-metallic fire check doors - Resistance test and Part – 2 performance criteria.

4.00 HOLLOW METAL FIRE DOOR WITH HONEY COMB CORE

4.01 Fire door shall be 2 hour fire rated and door quality shall be approved by TAC/CBRI and tested conformed to IS : 3614. The door should meet the requirements of Tariff Advisory Committee (TAC).

Unless otherwise specified, maximum size of door in this type :

Single shutter door	:	1150 mm x 3000 mm
Double shutter door	:	2000 mm x 3000 mm.

For doors of size 3000 mm height, the options shall be :

a) A man operation door of size above 2400 mm height shall be provided with a removable panel / fixed panel on top with glazing or without glazing as required.

b) The construction of above panel shall be designed similar to that of a shutter in case of flush panel to match the exteriors. In the case of glazing a 6 mm thick toughened glass with clipping arrangement on top shall be provided which shall be mounted on a transom of approved design.

4.02 Frame

a) **Material -** Frame to be manufactured from 1.60 mm (16 gauge) galvanized steel sheets complying with latest IS 277 coating class zinc coating mill phosphatized.

b) **Profile** - Door frame profile to be double rebated of dimensions 143 mm x 57 mm (+/- 0.3) with bending radius of 1.4 mm.

c) **Fabrication** - Frame to be fabricated from 1.60 mm thick galvanised steel sheet to the specified profile and dimensions. Frames fabricated at factory shall be in knock down form with butt joints for bolted assembly at site.

d) **Door frame preparations -** Frames to be provided with a 3 mm thick back plates on all jambs with provision for anchor bolt fixing to wall openings. All frames to have reinforcement pads for fixing of door closer, at appropriate location as per manufacturer's details.

e) Frames to have factory finish-pre-punched cut outs to receive specific hardware and iron mongery.

f) Frames to be provided with hinge plates 3 mm thick pre-drilled to receive hinges for screw mounted fixing. All cut outs including hinge plates, strike plates to have mortar guard covers from inside to prevent cement, dust ingress into cut outs at the time of grouting.

g) Frames to have rubber shutter silencer on strike jambs for single shutter frames and on the head jambs for double shutter frames.

h) **Finish:**
Door frames to be suitably cleaned with solvents and etch primered for receiving primer and top coats.

Door frames to be primered in zinc phosphate stoving primer (35 microns DFT).

Door frames to be finished in thermo setting paint (35 microns DFT) of approved colour and make as specified by KEIPL/PM.

4.03 **Fire door shutter**

a) **Material -** Fire door shutter to be manufactured from 1.25 mm (18 gauge) galvanized sheets conforming to latest IS : 277 coating class zinc coating, mill phosphatized.

b) **Fabrication -** Shutters to be press formed to 46 mm thick double skin hollow door with lock seam joints at stile edges. Shutters to have no visible screws or fasteners on either face. Internal reinforcement to be provided at top, bottom and stile edges for desired fire rating.

c) **Door shutter cores** - Shutters to be provided with honeycomb paper core to be bounded to the inner faces of the shutter.

d) **Door shutter preparations -** Shutters to be factory prepared with pre-punched cutouts and reinforcements to receive iron mongery as per final finish hardware schedule. The shutter should have an interlocking arrangement at this stile edges for flat surface on either side.

Shutters to have pre-drilled hinge plates with hinge guard covers. Shutters with locks to have concealed lock box with lock fixing brackets with pre-tapped holes.

e) For shutter with door closer reinforcement pads to be provided at appropriate location as per manufacturer's design.

All iron mongery preparation to have adequate reinforcement for flushes fixing at site.

f) **Vision panel for Fire rated door :** Vision panel to be provided with Boroslicate clear float glass of the thickness 6 mm for upto two hours fire rating. Glass to be fixed with clip on frames for square and rectangular vision panels and with spin turned rings for circular vision panels. Unless otherwise specified, standard sizes are 200 mm x 200 mm, and 380 mm diameter.

g) **Finish:** Shutters to be suitably cleaned with solvents and etch primered for receiving primer and top coats.

Shutters to be primered in zinc phosphate stoving primer (35 microns DFT).

Shutters to be finished in thermo setting paint (35 microns DFT) of approved colour and make as specified by KEIPL/PM.

4.04 **PACKING**

a) **Frame:** Individual frames members to be wrapped in polyethene sheets and placed in individual card board boxes. Individual boxes to be sealed. Frames to be assembled at site with aid of roofing bolts.

b) **Shutters:** Shutters to be wrapped in polyethene sheets and further packed in card board and duly strapped. All frames and shutters duly marked as per door schedule for easy identification at site.

4.05 **STORAGE**

All knocked down frames shall be stacked flat and shutters vertically on wooden runners and suitably covered as per the instructions of manufacturer to prevent rust and damage.

4.06 **INSTALLATION**

i. **Door frame fixing :** The door frames should be assembled adjacent to the place of installation as the frames are not designed for transporting in an assembled condition. After assembly it is to be ensured that all threaded preparations are covered from the back of the frame using 15 x 10 self adhesive sponge strip to prevent penetration of mortar back- fill into screw threads. The head member of assembled frame shall be positioned against jambs ensuring correct alignment and secured using M8 x 20 long plated bolts together with nuts spring and flat washers.

The assembled frame shall be kept in position within the opening by means of bracing. In order to correctly position the frame against finished floor level or equalise on adjustable floor anchors where specified, shim shall be used under jambs. The frame shall be checked for squareness, alignment, twist etc. with carpenters bevel and plumb.

A tie rod shall be fixed to the frame during installation to ensure the correct dimensions between the frame rebates and the same may be removed after installation.

Where a 2nd fix application is required a shim detail is suggested to take up gap between frame and existing opening.

Existing masonry wall openings - Metal expansion shields

a) Brace, position, level etc.

b) Mark all positions of fixings on wall.

c) Remove frame and drill wall to appropriate specified size.

d) Fit rod anchor shells metal expansion bolts into the wall.

e) Fit jamb spacer bracket into back of frame profile.

f) Reposition frame back into opening and realign.

g) Lightly screw CSK HD machine screws into shells, shim behind frame.

h) Slowly tighten screws continually checking plumb, square etc. Finally ensure frames are not deformed as tightened.

i) After fixing the frame shall be grouted with cement mortar 1:3 or Plaster of Paris or Gypsum powder as approved. Gap between frame and wall to be closed by cement pointing using cement mortar 1:3.

j) Back fill the frame through holes provided and insert nylon plugs.

ii) **Door shutter fixing:** Fix all the hardware to the door shutter like hinges, flush bolts, bolts, mortice locks, door closer, door stoppers, handles etc. with the appropriate screws and bolts supplied.

The shutter is to be then fixed to the frame which is already installed. Align the shutter to match the hardware to the cutouts in the frame. Tighten the hinge screws.

iii) Application of 'FLAT' seal (for smoke check if specifically required)

Clean door jamb rebate surfaces of all dust, oil etc. Affix self-adhesive 'FLAT' seal on the door frame rebates as indicated by the manufacturer on hinge jambs, strike jambs, head member and sill.

4.07 **FIRE DOOR - HARDWARE SCHEDULE**
Hardware Details as per BOQ Specification

5.00 **HOLLOW GENERAL PURPOSE DOORS WITH HONEYCOMB CORE**

5.01 Unless otherwise specified, maximum size of door in this type :

Single shutter door : 1150 mm x 3000 mm

Double shutter door : 2000 mm x 3000 mm

For doors of size 3000 mm height, the options shall be :

a) A man operation door of size above 2049 mm height shall be provided with a removable panel / fixed panel on top with glazing or without glazing as required.

b) The construction of above panel shall be designed similar to that of a shutter in case of flush panel to match the exteriors. In the case of glazing a 6 mm thick toughened glass with clipping arrangement on top shall be provided which shall be mounted on a transom of approved design.

5.02 **FRAME**

a) **Material -** Frameto be manufactured from 1.25 mm (18 gauge) galvanised steel sheets complying with latest IS 277 coating class zinc coating mill phosphatized.

b) **Profile -** Door frame profile to be single rebated of dimensions 100 mm x 57 mm (+/- 0.3) with bending radius of 1.2 mm.

c) **Fabrication -** Frame to be fabricated from 1.25 mm thick galvanised steel sheet to the specified profile and dimensions. Frames fabricated at factory shall be in knock down form with butt joints for bolted assembly at site.

d) **Door frame preparations -** Frames to be provided with a 3 mm thick back plates on all jambs with provision for anchor bolt fixing to wall openings. All frames to have reinforcement pads for fixing of door closer at appropriate location as per manufacturer's details.

e) Frames to have factory finish - prepunched cut outs to receive specific hardware and iron mongery.

f) Frames to be provided with hinge plates 3 mm thick pre-drilled to receive hinges for screw mounted fixing. All cut outs including hinge plates, strike plates to have mortar guard covers to prevent cement, dust ingress into cut outs.

g) Frames to have rubber shutter silencer on strike jambs for single shutter frames and on the head jambs for double shutter frames.

h) **Finish -** Door frames to be suitably cleaned with solvents and etch primered for receiving primer and top coats.

Door frames to be primered in zinc phosphate stoving primer (35 mircons DFT).

Door frames to be finished in thermo setting paint (35 microns DFT) of approved colour and make as specified by KEIPL/PM.

5.03 **GENERAL PURPOSE DOOR SHUTTER**

a) **Material :** General purpose door shutter to be manufactured from 0.80 mm (22 gauge) galvanised sheets conforming to latest IS : 277 coating class zinc coating, mill phosphatized.

b) **Fabrication:** Shutters to be press formed to 46 mm thick double skin hollow door with lock seam joints at stile edges. Shutters to have no visible screws or fasteners on either face.

c) **Door shutter core :** Shutters to be provided with honeycomb paper core to be bounded to the inner faces of the shutter.

d) **Door shutter preparations:** Shutters to be factory prepared with pre-punched cutouts and reinforcements to receive iron mongery as per final finish hardware schedule. The shutter should have an interlocking arrangement at this stile edges for flat surface on either side.

Shutters to have pre-drilled hinge plates with hinge guard covers. Shutters with locks to have concealed lock box with lock fixing brackets with pre-tapped holes.

All iron mongery preparation to have adequate reinforcement for flush fixing at site.

e) For shutter with door closer reinforcement pads to be provided at appropriate location as per manufacturer's design.

f) **Vision panel:** Vision panel to be provided with toughened float glass of the thickness 6 mm. Glass to be fixed with clip on frames for square and rectangular vision panels and with spin turned rings for circular vision panels. Unless otherwise specified, standard sizes are 130 mm x 270 mm, 200 mm x 300 mm, 450 mm x 750 mm and 360 mm diameter.

g) **Finish:** Shutters to be suitably cleaned with solvents and etch primered for receiving primer and top coats.

Shutters to be primered in zinc phosphate stoving primer (35 microns DFT).

Shutters to be finished in thermo setting polyurethane aliphtic grade paint (35 microns DFT) of approved colour and make as specified by KEIPL/PM.

5.04 **PACKING**

a) **Frame;** Individual frames members to be wrapped in polyethene sheets and placed in individual card board boxes. Individual boxes to be sealed. Frames to be assembled at site with aid of roofing bolts.

b) **Shutters:** Shutters to be wrapped in polyethene sheets and further packed in card board and duly strapped. All frames and shutters duly marked as per door schedule for easy identification at site.

5.05 **STORAGE**

Storage shall be the same as specified in clause 4.05 for fire door.

5.06 **INSTALLATION**

Installation shall be the same as specified in clause 4.06 for fire door.

5.07 **GENERAL PURPOSE DOOR - HARDWARE SCHEDULE**

Hardware Schedule as per BOQ Specification

6.00 **FIELD INSPECTION**

After installing the door, the Contractor shall test the performance of the shutter in the presence of the KEIPL/PM. The doors shall be smoothly operable under all ambient conditions. All control and locking devices shall give fault free performance.

7.00 **MODE OF MEASUREMENT**

Measurement shall be in square metre for clear area over one face inclusive of exposed frame. Portions embedded in masonry / concrete / flooring shall not be measured.